Korean air flight 801

Flight model: Boeing 747
Date: August 06, 1997

From: Seoul, South Korea
To: Guam, US territory

The weather near Guam is very rainy. The glide slope beacon in Guam airport is also removed for maintains. The captain had another flight scheduled but, in the last minute schedule change he had to take this plane. He captain feels exhausted.

Problem

Clouds and rain also, block the visual view of runway. The pilots change the plane's path in order to gain a clear visual view. In the middle of the landing the crew, get a glide slope warning which make the crew confused. As the crew approaches the airport, the crew could hardly see the runway. Then, the ground proximity warning pops up telling the crew they are 500 feet above the ground. Just, above 200 meters, unable to see the runway the first officer make request to go around. But, the pilot due to fatigue, took the decision late making it too late for a go around procedure.

Investigation

When the glide slope is not working the crew must follow a approach called as **Step down approach** where the pilot steps down certain altitude after a every pre-fixed points. All theses are given in the crew's charts including the altitude to descend and point at which to descend. Using the FDR, it is found that the crew makes the first step correctly but failed to make the subsequent steps rather they made the plane to descend in a slope. The sudden working of the glide slope made the pilot distracted. After accident, many survivors legs got stuck with the cross bar of the seat make them unable to move. One of the survivor who is a flight engineer also tells that after crash the flight engulfed in fire probably due to the presence of duty free alcohol which was stored in the cabin above along with the oxygen. It was found that the crew was provided with a landing chart which was 6 months old and out off date. The Korean air had trained their pilots to land with the aid of **DME (Distance measuring equipment)**, Which tells the pilots how far they are from the airport. Usually the DME beacon that relay signal for DME is kept at the foot of the runway. But, in Guam it is kept 5 KM away from the runway. The crew expected to the DME would guide them to the airport runway. Also it was found that the FAA, recommends planes to turn off the **MSAW (Minimum safe altitude warning)**, The MSAW sends a warning to the ATC when the plane are flying to low. The MSAW in Guam gave nuisance reading and were extended to open sea where the system were of no use. The MSAW tracked the planes which were 80KM away from the airport which is a open sea.

Changes that were brought after the incident

Reports point airline for inadequate training and crews over reliance on automation. The report also points the FAA for its negligence on the MSAW. The flight engineer still argues for better policy on the flight alcohol stored in the cabin and better seat design.

Reference

1. Worst Equipment Failures in Aviation History — Mayday: Air Disaster (00:00 - 49:00)
   `https://youtu.be/_aIR6BLdZ7Q`